

Restock Bathroom Caddy SOP

One episode is one full restock of a bathroom caddy on the tabletop station. It runs from a dirty empty caddy, three toiletries standing scrambled on a staging tray, a soap pump bottle low on soap in its fill stand, and two hand towels lying flat on a towel mat, to a wiped caddy holding two rolled towels stacked on its towel shelf and four items standing upright over their own labels on its item shelf, with the pump filled to its fill line and its head seated.

The caddy has **two open shelves and nothing else**: no bays, no dividers between items, no walls to drop an item into. The towel shelf at the back holds the stacked rolls, the item shelf at the front holds the four upright items, and a low rail separates them. Nothing on either shelf holds an item up, so every item is stood down on its own base, clear of its neighbours, and left standing.

Arm roles are fixed and never swap. The **right gripper is the item gripper**: the toiletries, the pump bottle, the pump head, the refill bottle, and the towels. It does every pick, every pour, and every placement, and it works the staging tray, the fill station, the towel mat, and the caddy. The **left gripper is the cloth gripper**: it takes the cloth, wipes the caddy, and parks the cloth. Its one exception is Step 3.1, where a hand towel is too wide for one gripper to wind straight, so both grippers roll it together. That is the only shared action in the task. Nothing is ever handed from one gripper to the other, and only one arm is over the caddy at a time.

Items live on the right side of the table and the cloth lives on the left, and neither arm reaches into the other arm's side. Two zones straddle the table midline so either arm reaches them without extending: the **caddy** at the center back, entered by one arm at a time, and the **towel mat** at the center front, worked by both grippers during the roll.

The order never changes: **wipe both shelves, refill the soap pump, roll and stack the two towels, then load the item shelf left to right**. The caddy is wiped before anything goes onto it, the towel shelf at the back is filled before the item shelf at the front, and the item shelf is loaded left to right so nothing is carried over an item already placed.

Never move or lift the caddy. Never put anything onto a shelf that has not been wiped, and never move more than one item per trip. Never drag or slide an item: lift it clear of the surface and carry it. Never turn an item in the air to change the way it faces: stand it down and take it again. Never turn the pump head, never lift the pump bottle by its head, and never lift it out of the fill stand before the head is seated. Never grip a toiletry by its cap or lid. Never set the cloth down anywhere but its park. Never fold a towel. Never change which item belongs in which slot, and never lean one item against another to keep it standing.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the caddy and its frame includes the caddy, the cloth park, the staging tray, the fill station, and the towel mat.
3. The dirt on both shelves, the four slot labels, and the fill line on the pump bottle are all readable on camera.
4. Both arms are at home with grippers open.
5. The table holds nothing but the caddy, the cloth on its park, the staging tray and its three toiletries, the fill station and its three pieces, and the towel mat and its two towels.
6. The left arm reaches both shelves, the cloth park, and both towel lanes without extending to a joint limit, and reaches nothing else on the table.
7. The right arm reaches the staging tray, all three fill station spots, both towel lanes, and the whole caddy without extending to a joint limit, and reaches nothing on the left side of the table.
8. Each arm comes onto both shelves straight down from above and from the front, without reaching past the back table edge. Nothing overhangs either shelf.
9. Arm paths to the caddy and the towel mat are clear, neither arm collides with the caddy on the way into the center, and both grippers stand side by side on one towel lane without the arms fouling each other. If a zone cannot be reached, move it toward the arm that cannot reach it until lines 6 to 8 hold.

Materials checklist

1. The caddy stands empty, level, and steady on the table, squared to the back table edge, and does not slide under a firm wipe.
2. The caddy has exactly two flat open shelves split by the low divider rail, no walls or bays anywhere else, and nothing above either shelf. The rail stands lower than the shortest item, so every drop onto either shelf is straight down.
3. Both shelves are empty and carry visible dirt.
4. The item shelf is deep enough front to back that every item stands fully on it with clear space in front and behind, and long enough left to right that the four items stand over their labels without touching each other.
5. Four slot labels run left to right along the front face of the caddy, each readable on camera: shampoo bottle, conditioner bottle, deodorant stick, soap pump. The labels are the only thing dividing the item shelf.
6. The staging tray holds the shampoo bottle, the conditioner bottle, and the deodorant stick standing upright in a row in a scrambled order that does not match the slot order, each with its label facing the front of the table.

7. The three toiletries are visibly different in shape, stand upright unaided on a bare flat surface, are closed and dry, and can each be lifted by one gripper on the body clear of the cap or lid.
8. The fill station holds three fixed spots in a row: the **fill stand** at the left, the **head rest** in the middle, and the **refill spot** at the right. The head rest is empty with its hole clear.
9. The soap pump bottle stands in the fill stand, held upright and steady so it cannot tip, with its head seated loose in the neck so it lifts straight out and presses straight back in without any turn. It stands upright unaided on a bare flat surface when full to its fill line and does not rock when set down.
10. The soap stands well below the fill line, and the line is visible from the front.
11. The refill bottle stands upright on the refill spot, open at its pour spout with no cap on, and holds more soap than one fill needs.
12. The towel mat sits centered at the front of the table, directly in front of the caddy, with its left lane and its right lane straddling the table midline.
13. The two hand towels lie fully flat on the mat, one per lane, each square with its near edge parallel to the front table edge, not overlapping and not overhanging the mat.
14. Both towels are the same size, are dry, are wide enough that both grippers stand on the near edge without touching, and hold a roll without springing open. A finished roll fits fully on the towel shelf behind the divider rail without overhanging the caddy at either end.
15. The folded cloth lies flat and damp on the cloth park, and the park holds nothing else.

Workspace layout

There is no wall behind the station, and nothing is marked or taped out apart from the slot labels on the caddy. Every area below is a fixed place on the bare tabletop, judged by eye against the table edges and the caddy itself.

- **Back table edge:** the far edge of the table. The caddy is squared to it and nothing overhangs it.
- **Caddy:** the open topped organizer in the center of the table, just inside the back table edge. It carries its two shelves side by side on one level, split by the divider rail. Nothing sits above either shelf, so both are worked straight down from above. It is never lifted, slid, or turned.
- **Towel shelf:** the flat open shelf across the back of the caddy, between the back wall and the divider rail. It holds the two rolled towels stacked, and is filled before the item shelf.
- **Item shelf:** the flat open shelf across the front of the caddy, between the divider rail and the front edge. It holds the four items standing upright over their labels. It has no walls or bays.
- **Divider rail:** the low rail between the two shelves. Nothing rests on it. Rolls stay behind it, items stay in front of it.
- **Slot:** one of the four places on the item shelf, numbered 1 to 4 left to right, each marked by its label on the front face of the caddy. A slot is a place on an open shelf, not a bay, so an

item standing in it is held up by nothing but itself.

- **Cloth park:** the area at the left end of the table where the folded cloth rests. Left gripper only.
- **Staging tray:** the tray to the right of the caddy holding the three toiletries at episode start, empty at episode end. Right gripper only.
- **Fill station:** the run of three spots to the right of the caddy where the soap is refilled. The **fill stand** holds the pump bottle upright and steady, the **head rest** holds the pump head upright by its dip tube, and the **refill spot** holds the refill bottle. The fill stand and the head rest are empty at episode end. Right gripper only.
- **Towel mat:** the mat centered at the front of the table, directly in front of the caddy, holding the two flat towels at episode start, one in its **left lane** and one in its **right lane**. Both lanes straddle the midline so each arm reaches either lane without extending. Empty at episode end.
- **Arm home:** the resting pose for each arm, gripper open and clear of the table.

Vocabulary

- **Arm assignment:** fixed by object, not by where the object lies. The **right gripper** handles every item and works the staging tray, the fill station, the towel mat, and the caddy. The **left gripper** handles the cloth and works the cloth park and the caddy. Its one exception is the towel roll in Step 3.1. It touches no other item at any point, and nothing is handed between grippers.
- **Idle arm:** the arm not named in a step. It waits at its home, clear of the table, for the whole of that step. It never steadies an item, holds a towel flat, holds the cloth, or assists the working arm. **Step 3.1 is the only step with no idle arm.**
- **Rolling together:** both grippers hold the same towel and move as one. They take the near edge the same distance in from each corner, start and stop together, travel at the same speed, and stay the same distance apart, so the edge they carry stays parallel to the far edge of the towel. Neither gripper leads, drags, or works part of the towel on its own.
- **Item:** any object the right gripper handles: a toiletry, the pump bottle, the pump head, the refill bottle, or a towel.
- **Toiletry:** one of the three staged objects. The **shampoo bottle** is tall and flat sided, the **conditioner bottle** is shorter and round bodied, and the **deodorant stick** is a short capped cylinder. Each carries a label on one face.
- **Standing in its slot:** the item stands upright on its own base on the item shelf, over the label its own name is on, fully on the shelf with clear space in front and behind. It stands on its own: it does not lean on a neighbour, does not touch a neighbour, does not rest against the divider rail, and no part of its base hangs over the front edge. A **toiletry** also stands with its label facing the front of the caddy, carried over from the way it stood on the staging tray. The **soap pump** has no facing to set: its spout may point any way.

- **Fill line:** the line marked on the pump bottle. The bottle is filled when the soap surface reaches it. Soap above the line is overfilled, soap below it is underfilled.
- **Seated head:** the pump head sits square and fully down in the bottle neck, collar against the neck, standing upright rather than tilted. It is seated from directly above: lowered straight down the line of the neck and pressed home. It is never turned into place.
- **Spill:** any soap outside the pump bottle: on the fill stand, the refill bottle, the table, the caddy, or any item.
- **Roll:** a hand towel wound up along the mat into a cylinder that holds its shape and does not unwind after release. A roll is **square** when its two ends are level and its axis runs parallel to the front table edge. A towel folded flat instead of wound is not a roll.
- **Cloth:** the damp folded cloth used on the caddy, held so a broad flat side meets the surface, never by a corner and never balled up.
- **Wiped:** a shelf is wiped when no dirt is visible anywhere on it: end to end, back to front, and into both ends. **Dirt** is any visible mark, dust, crumb, smear, or residue. How many passes the cloth makes does not matter. What is judged is what is left behind.
- **Release point:** the right gripper opens only over the towel mat, a fill station spot, the towel shelf, or the slot the item belongs in. The left gripper opens only over the cloth park or over the towel lane it is rolling on. Neither ever opens over open table, over another item, or over a slot the held item does not belong in.
- **Stable placement:** the item stays still after release, fully inside its area, and does not rock, lean on a neighbour, roll, or slide.

Steps

Step 1: Wipe the caddy

Goal: both shelves are clear of dirt and the cloth is back on its park.

The right arm is the idle arm for the whole of Step 1. Wipe the **towel shelf first, then the item shelf**, so the back of the caddy is never reached across after the front is clean.

1.1 Take the cloth

- The **left gripper** approaches the cloth on its park from above and closes on it so it is held flat, with enough of it below the gripper to meet the surface.
- Lift it clear of the park and bring it to the caddy.

Check: the cloth presents a flat face and does not hang by a corner or slip in the gripper. If it does, set it down on the park and take it again.

1.2 Wipe the two shelves

- Set the flat face of the cloth down on the **towel shelf** and wipe it end to end, keeping the cloth flat and pressing only hard enough to lift the dirt.

- Go back over any mark that stays. Do not leave the shelf until no dirt is visible on it, including the strip against the back wall, the strip against the divider rail, and both ends.
- Lift the cloth clear of the divider rail, set it down on the **item shelf**, and wipe that shelf the same way, end to end and from the rail out to the front edge.

Keep the cloth flat and on the shelf while it is moving. Do not drag it over the divider rail, do not push it off the front edge, and do not press the caddy forward.

Check: no dirt is visible on either shelf. Do not start Step 2 with dirt anywhere on the caddy.

1.3 Park the cloth

- Carry the cloth to the cloth park, lay it down flat, release, lift clear, and return to home.

Check: the cloth lies fully on the park, not draped over the table or the caddy, and the left gripper is empty and at home before the right arm moves.

Step 2: Refill the soap pump

Goal: the pump bottle is filled to its fill line, its head is seated, and the fill station is tidy.

The left arm is the idle arm for the whole of Step 2. The pump bottle stays in the fill stand for this whole step and is never lifted.

2.1 Lift the pump head onto the head rest

- The **right gripper** approaches the pump head from above and closes on its collar, square to it so it hangs upright.
- Lift it straight up out of the neck without turning it, until the dip tube is clear of the bottle.
- Carry it to the head rest, lower it straight down so the dip tube drops into the hole, and release once the head stands upright on its own.

Check: the head stands upright on the rest and the bottle neck is open and clear. If it leans or will not stand, take it again and stand it once more.

2.2 Pour the refill into the bottle

- Approach the refill bottle from above, close on its body, and lift it straight up clear of its spot.
- Carry it over the open neck of the pump bottle and hold it with the spout over the neck.
- Tip it slowly until soap runs into the neck, and keep pouring until the soap reaches the fill line.
- Bring the refill bottle back upright over the neck before moving it away, so the last of the soap runs off the spout into the bottle.
- Carry it back to the refill spot, set it down upright, and release.

Keep the spout over the neck the whole time. Do not pour from high above the bottle, do not pour past the fill line, and do not rest the spout on the neck rim.

Check: the soap reaches the fill line, there is no spill anywhere, and the refill bottle stands upright on its spot. If the fill stopped short of the line, take the refill bottle again and pour once more.

2.3 Seat the pump head

- Close on the head's collar on the head rest and lift it straight up until the dip tube is clear of the hole.
- Carry it across at that height until it hangs directly above the bottle neck, dip tube lined up on the opening.
- Lower it straight down the line of the neck until the dip tube is in, then press straight down until the collar meets the neck.
- Release and lift clear.

The head is seated from above and from above only. Do not turn the head or the wrist to line it up, do not swing it in from the side, and do not push it down at an angle.

Check: the head is seated. If it is tilted or standing proud, lift it straight up out of the neck and lower it in again once from directly above. Its spout direction is not checked and is never twisted to change it.

Step 3: Roll and stack the two hand towels

Goal: both towels are rolled and stacked on the wiped towel shelf, and the towel mat is empty.

Work the **right lane towel first, then the left lane towel**, finishing one before starting the other. **Step 3.1 uses both grippers together. Step 3.2 uses the right gripper alone**, with the left arm back at its home.

3.1 Roll the towel with both grippers

- Bring both grippers down onto the near edge of the towel from above: the **left gripper** near its left corner, the **right gripper** near its right corner, each the same distance in from its corner. Close both.
- Lift the near edge just clear of the mat and lay it forward onto the body of the towel, both grippers moving together, so the towel takes a soft first turn across its full width rather than a flat crease. Keep the edge parallel to the far edge the whole way over.
- Open both grippers, close them empty, and rest their closed tips on the turn, one near each end.
- Push the turn away from you along the lane in one continuous run to the far edge, both grippers travelling together, so the towel winds up between them into a roll that stays square.
- With both grippers still closed, press lightly along the roll once, then lift both clear.
- Return the **left arm** to its home before the roll is lifted. It takes no further part in Step 3.

Roll the towel where it lies, inside its own lane. Do not lift it off the mat to wind it in the air, do not pull it across the table, and do not fold it flat instead of rolling it.

Check: the towel is a square roll, lying inside its lane, with no loose flap hanging off it. If it unwinds, comes out crooked, or a flap hangs loose, lay it flat and square on its lane and roll it again once with both grippers.

3.2 Place the roll on the towel shelf

The left arm is the idle arm for the whole of Step 3.2, so only one arm goes over the caddy.

- The **right gripper** closes across the middle of the roll, square to it so the roll hangs level, and lifts it straight up clear of the mat.
- Carry it to the towel shelf and lay it down with its length running left to right, parallel to the caddy's back wall.
- For the **first roll**, lay it centered on the shelf, fully on the shelf, clear of the back wall and behind the divider rail.
- For the **second roll**, lay it centered on top of the first, running the same way, so the two rolls sit square above each other.
- Release once the roll rests still, then lift clear.

Do not carry a roll over the item shelf, do not rest one on the divider rail, and do not drop one onto the shelf from above.

Check: the roll rests still and holds its shape, sits fully on the towel shelf or fully on the roll below it, stays behind the divider rail, and does not overhang the caddy at either end. If it rolls, unwinds, overhangs, or comes to rest on the rail, lift it clear, lay it on its lane, and roll and place it again once. After both towels, the towel mat is empty and two rolls sit stacked and square.

Step 4: Load the item shelf

Goal: all four slots hold their own item, each standing upright and on its own over its label.

Work the item shelf **left to right**, slot 1 first, one item per trip, with the **right gripper** only. Read the slot label before each pick and take the item that label names. The left arm is the idle arm for the whole of Step 4. Nothing on this shelf holds an item up, so set each item down on its own base, clear of the item already placed to its left, and never close the gap by leaning one against another.

4.1 Place a toiletry from the staging tray, slots 1 to 3

- Read the label for the slot you are filling and find that toiletry on the staging tray.
- Approach it from above and close on its body, square to it so it hangs upright, clear of its cap or lid, with the label facing you.
- Lift it straight up clear of the tray and carry it to a point directly above its own label, keeping the label facing you for the whole carry and keeping the item clear of the divider rail and of

any item already placed.

- Lower it straight down until it stands on the shelf on its own base, hold it a moment until it is clearly standing on its own, then open the gripper and lift clear.

If it comes up facing the wrong way, stand it back on the staging tray, let go, and take it again with the label facing you. Never turn it in the air. Do not let go until the item is taking its own weight, and do not steady it after release.

Check: the item is standing in its slot as Vocabulary defines it, label facing the front. If it is in the wrong slot, lift it clear, stand it on the staging tray, and place it again over the label its own name is on. If it leans, touches a neighbour, or sits over the front edge, lift it clear and stand it down again once.

4.2 Place the soap pump bottle in slot 4

- Approach the pump bottle in the fill stand from above and close on its body below the shoulder, square to it so it hangs upright, clear of the head.
- Lift it straight up out of the fill stand, keeping it level so no soap runs out of the spout.
- Carry it to a point directly above the soap pump label and lower it straight down until it stands on the shelf on its own base.
- Hold it a moment until it is clearly standing on its own, then open the gripper and lift clear.

The pump has no facing to set, so carry it the way it comes out of the stand and set it down that way. Never lift it by its head, tip it during the carry, or turn it in the air. A full bottle is top heavy, so lower it slowly and let it settle before releasing.

Check: the bottle is standing in slot 4 as Vocabulary defines it, its head still seated square, and no soap is on the bottle, the caddy, or the table. If it rocks or leans, lift it clear and stand it down again once.

Check: all four slots hold their own item, no two items touch, and the staging tray, the fill stand, and the head rest are empty.

Step 5: End the episode

- Confirm the wipe: no dirt is visible on either shelf.
- Confirm the towels: two rolls sit stacked and square behind the divider rail, holding their shape and not overhanging the caddy.
- Confirm the item shelf: all four items are standing in their slots, each toiletry with its label facing the front.
- Confirm the soap: the bottle holds soap to its fill line, its head is seated, and no soap is anywhere outside the bottle. Its spout may point any way.
- Confirm the zones: the staging tray, the towel mat, the fill stand, and the head rest are empty, the refill bottle stands upright on its spot, and the cloth lies flat on its park.

- Correct any failed check before ending.
- Return both arms home with grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the four items off the item shelf. Stand the three toiletries on the staging tray in a scrambled order that does not match the slot order, each with its label facing the front of the table. Stand the pump bottle in the fill stand with its head seated loose in the neck.
2. Pour soap out of the pump bottle back into the refill bottle until the level stands well below the fill line, then top the refill bottle up so it holds more than one fill needs. Wipe any soap off the bottles and the fill station.
3. Take both rolls off the towel shelf, unroll them, and lay them flat and square in their lanes, near edge parallel to the front table edge, not overlapping and not overhanging.
4. Put dirt back on both shelves so the next episode has something to wipe.
5. Tighten any toiletry cap that has loosened, and replace a toiletry that will not stand upright unaided on a bare flat surface.
6. Replace a towel that will no longer hold a roll, and a pump head that no longer seats with a straight press.
7. Rinse or replace the cloth, fold it flat and damp, and lay it on the cloth park.
8. Check the caddy is still squared to the back table edge and still does not slide, then clear any debris off the table.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** the steps run out of order, the item shelf is wiped or filled before the towel shelf, the item shelf is loaded other than left to right, the **left gripper** parks the cloth before both shelves are wiped, the **right gripper** lifts the pump bottle out of the fill stand before its head is seated, an item or a roll is carried over an item already standing on the item shelf, or a step is skipped.
- **SOP rule broken:** Steps 1 to 4, wipe both shelves, refill the pump, roll and stack the towels, load the item shelf left to right, and carry nothing over an item already placed.
- **Coaching note:** wipe, refill, towels, items. Towel shelf before item shelf, items left to right, and come in over empty space.

Violation: Arms swapped roles

- **Visible cue:** the **left gripper** touches a toiletry, the pump bottle, the pump head, the refill bottle, or a towel outside Step 3.1; the **right gripper** touches the cloth; a step that names a gripper is done with the other one; the left gripper reaches past the caddy to the staging tray or the fill station; or the right gripper reaches past the caddy to the cloth park.
- **SOP rule broken:** Steps 1 to 4, the right gripper handles every item and the left gripper handles the cloth, and neither arm reaches into the other arm's side.
- **Coaching note:** right gripper handles items, left gripper handles the cloth, each arm works its own side. The towel roll in Step 3.1 is the one shared action.

Violation: Idle arm interferes

- **Visible cue:** outside Step 3.1, the arm not named in a step enters the caddy, steadies an item, holds a towel flat, holds the cloth, or assists the working arm; both arms are over the caddy at once; or the **left arm** stays on the towel mat during Step 3.2 instead of going home.
- **SOP rule broken:** Steps 1 to 4, only one arm is over the caddy at a time and the arm not named in a step waits at its home.
- **Coaching note:** one arm works, the other waits at home. The left arm goes home before the roll is lifted.

Violation: Shelf left with dirt, or something placed on dirt

- **Visible cue:** dirt is still visible on a shelf when the episode moves on, a shelf is never touched by the cloth, an area is never reached (the strip against the back wall, the strip against the divider rail, an end of a shelf, or the front of the item shelf), or the **right gripper** lays a roll or stands an item on a shelf that still shows dirt.
- **SOP rule broken:** Steps 1.2 and 4, a shelf is wiped only when no dirt is left on it, and nothing goes onto a shelf that has not been wiped.
- **Coaching note:** the shelf is done when nothing is left on it, not when the cloth has been over it once. Anything placed on dirt comes back off and the shelf is wiped again.

Violation: Cloth held or set down wrong

- **Visible cue:** the **left gripper** holds the cloth by a corner, balled up, or edge on so only a narrow line touches the surface; lays it on the table, a shelf, the towel mat, or an item instead of on its park; or still holds it when Step 2 starts.
- **SOP rule broken:** Steps 1.1 to 1.3, the left gripper holds the cloth flat and sets it down nowhere but its park.
- **Coaching note:** a broad flat face on the surface, and the cloth only ever rests on its park.

Violation: Pump head mishandled

- **Visible cue:** the **right gripper** turns the head or its wrist to line the head up or to set the spout direction, swings the head in from the side instead of lowering it from directly above, pushes it in at an angle, leaves it tilted or standing proud at the end of Step 2, lays it on the table or the fill stand instead of standing it on the head rest, or drags its dip tube across a surface.
- **SOP rule broken:** Steps 2.1 and 2.3, never turn the pump head, and leave it seated square on the neck.
- **Coaching note:** straight up to remove, straight onto its rest, straight down from above to seat. The head never turns, and the spout points wherever it lands.

Violation: Fill line missed

- **Visible cue:** the **right gripper** stops the pour with the soap visibly short of the fill line and the episode moves on, or pours past the line so the soap stands above it.
- **SOP rule broken:** Step 2.2, fill the bottle to its fill line.
- **Coaching note:** watch the line rather than the refill bottle, and slow the pour as the level comes up.

Violation: Soap spilled

- **Visible cue:** soap runs down the outside of the pump bottle, drips onto the fill stand, the table, the caddy, or an item, or the **right gripper** tips the refill bottle away from the neck while it is still running.
- **SOP rule broken:** Step 2.2, pour with the spout over the neck and bring the refill bottle upright over the neck before moving it away.
- **Coaching note:** keep the spout over the neck from the first tip to the last drip.

Violation: Towel folded, lifted off its lane, or left loose

- **Visible cue:** the towel is creased flat into a folded rectangle or laid on the shelf flat rather than wound into a roll; it is lifted off the mat and wound in the air, pulled onto the bare table, or rolled across the other towel's lane; or the finished roll springs open, hangs a loose flap, or unwinds on the shelf and the episode moves on.

- **SOP rule broken:** Step 3.1, roll each towel where it lies inside its own lane, and leave it tight so it holds its shape.
- **Coaching note:** a soft first turn, then push it up along the lane and press along its length before it is lifted. A loose roll opens on the shelf.

Violation: Towel not rolled by both grippers together

- **Visible cue:** one gripper works the towel while the other waits at home or hovers without taking it, the second gripper comes in only part way through the roll, one gripper leads and the other trails, they take the near edge at visibly different distances from their corners, they drift apart or together during the run, or the edge goes over skewed so the finished roll is crooked.
- **SOP rule broken:** Step 3.1, the left gripper and the right gripper roll the towel together.
- **Coaching note:** both ends of the near edge, taken at the same time and matched before either arm moves. The two arms start, travel, and stop as one.

Violation: Towel stack wrong

- **Visible cue:** the **right gripper** lays the second roll beside the first instead of on it, the rolls lie across each other rather than parallel, a roll overhangs the caddy, a roll comes to rest on the divider rail, or a roll ends up on the item shelf.
- **SOP rule broken:** Step 3.2, lay the second roll centered on top of the first, running the same way, fully on the towel shelf and behind the rail.
- **Coaching note:** both rolls run left to right, one squarely on the other, both behind the rail.

Violation: Item in the wrong slot

- **Visible cue:** the **right gripper** stands an item over a label naming a different item, or two items end up swapped between slots.
- **SOP rule broken:** Step 4, read the slot label and take the item it names.
- **Coaching note:** read the label, then pick. The tray order is scrambled on purpose.

Violation: Item not standing on its own in its slot

- **Visible cue:** an item leans, rocks, or is left tilted; its base sits over the front edge; it sits visibly off its own label and into a neighbour's space; it rests against the divider rail; two items touch or one only stays upright because a neighbour props it; or a toiletry stands with its label away from the front.
- **SOP rule broken:** Step 4, each item stands on its own base over its own label, clear of its neighbours.
- **Coaching note:** every item stands on its own base over its own label with clear space beside it. Nothing on this shelf holds it up.

Violation: Released before the item was standing

- **Visible cue:** the **right gripper** opens while the item is still tilted, still swinging, or not yet taking its own weight, and the item then rocks, leans, or topples.
- **SOP rule broken:** Step 4, hold the item a moment until it is clearly standing on its own, then open the gripper.
- **Coaching note:** let it settle on its base first. On an open shelf the release is the moment the placement is won or lost.

Violation: Item turned in the air

- **Visible cue:** an item is lifted and rotated in the gripper to change the way it faces, or the pump head or the pump bottle is spun mid carry.
- **SOP rule broken:** Steps 2 and 4, never turn an item in the air.
- **Coaching note:** the gripper cannot reorient an item in the air. Stand it down, let go, and take it again with the facing wanted.

Violation: Item gripped by its cap, lid, or head

- **Visible cue:** the **right gripper** closes on a toiletry cap or lid, or on the pump head, to lift the item it belongs to.
- **SOP rule broken:** Steps 4.1 and 4.2, never grip a toiletry by its cap or lid and never lift the pump bottle by its head.
- **Coaching note:** grip the body. An item carried by its cap can come apart mid carry.

Violation: Caddy moved

- **Visible cue:** the caddy slides, turns, or is lifted during the episode, whether it is pushed by an arm, dragged by the cloth, or nudged by an item going down.
- **SOP rule broken:** Steps 1 to 4, never move or lift the caddy.
- **Coaching note:** the caddy is the fixed part of this task. Ease off the wipe pressure and come onto the shelves straight down.

Violation: Bad pick, carry, or release

- **Visible cue:** an item is pushed, slid, or scraped across the table, the tray, the mat, or a shelf instead of being lifted clear and carried; two or more items are carried in one trip; the refill bottle is set down leaning, on its side, or off its spot, or is left in the **right gripper** while another action starts; a gripper opens over open table, over another item, or over a slot the held item does not belong in; or an item is dropped from above rather than set down.
- **SOP rule broken:** Steps 2 to 4, one item per trip, lifted clear of the surface and carried, and released only over the spot it is going to.
- **Coaching note:** the item leaves the surface before it moves, and where it lands is known before the gripper opens.

Violation: Placement shoved or fiddled with

- **Visible cue:** a gripper nudges, pats, or slides an item, a roll, or the pump head into position after releasing it instead of lifting it clear and standing it down again; a gripper closes on an item or the cloth, finds the grip off, and closes again more than twice before lifting; or more than about two small corrections are made to settle a placement. The recoveries written in Steps 3 and 4 do not count.
- **SOP rule broken:** Steps 1 to 4, correct a bad placement by lifting the item clear and standing it down again, in one or two small corrections.
- **Coaching note:** lift it and stand it again rather than shoving it. If the first placement lands wide every time, the approach needs tightening rather than correcting after.

Violation: Required check not followed

- **Visible cue:** a failed check is visible and the episode continues without the written correction, or a recovery the SOP does not write is used.
- **SOP rule broken:** Steps 1 to 4, perform each check and use only its stated correction.
- **Coaching note:** stop at each check and use only its written recovery.

Violation: Dropped, collided, or knocked over

- **Visible cue:** an item, a towel, or the cloth falls or is struck; a toiletry topples on the tray or the shelf; the pump bottle tips in its stand; an arm knocks a placed item on approach or retract; or an item is pushed off a shelf or off the table.
- **SOP rule broken:** Steps 1 to 4, keep each item on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before the arm moves, and retract straight up so the gripper does not catch a standing item.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with dirt on either shelf, an empty slot, an item left on the staging tray or the towel mat, two items touching, soap short of or above the fill line, the pump head unseated, the cloth off its park, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 5, confirm the wipe, the towels, the item shelf, the soap, and the zones, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).

- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective toiletry:** one that will not stand upright unaided on a bare flat surface, slips out of a correct grip, or has a cap that will not stay tight.
- **Defective soap pump:** a head that will not seat under a correct straight press or binds under a correct straight lift, or a bottle that will not stand upright unaided when full to its fill line.
- **Defective refill bottle:** a spout that will not pour, runs so fast the line cannot be caught, or drips after a correct upright return.
- **Defective towel:** one so stiff, slick, or worn that it will not hold a roll after a correct two gripper roll and press.
- **Defective cloth:** one that is torn, shedding, dry at episode start, or so loaded with dirt that a correct wipe only smears it.
- **Caddy will not stay put:** it slides or turns under a correct wipe or a correct straight down placement.
- **Shelf will not hold an item:** a shelf that is not level, or so slick or so shallow front to back that a correctly stood item will not stay standing on it.
- **Damaged shelf surface:** a stain, scratch, or burn in the shelf itself that does not lift under a correct wipe.
- **A zone shifts during the episode,** or a zone turns out to sit outside its arm's comfortable reach so a shelf, a slot, or a fill station spot cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Take the cloth from the cloth park
2. Wipe the two shelves
3. Return the cloth to the cloth park
4. Lift the pump head onto the head rest
5. Pour the refill into the soap pump bottle
6. Seat the pump head in the soap pump bottle
7. Roll one hand towel
8. Place one rolled towel on the towel shelf
9. Place one toiletry in its slot
10. Place the soap pump bottle in its slot
11. Return both arms home and end the episode